higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

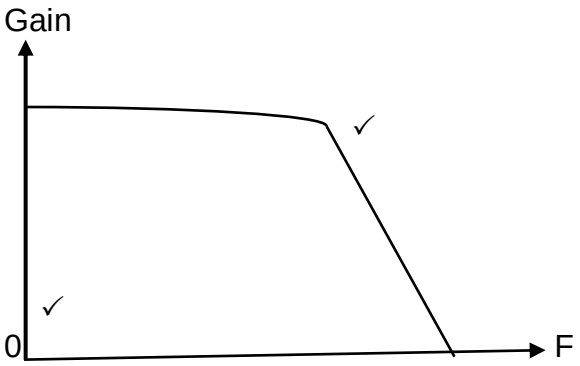
1 August 2024

This marking guideline consists of 9 pages.

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QUESTION 1

1.1



(2)

1.2 1.2.1 $X_L = 2\pi \times 1 \times 10^6 \times 120x10^{-6} = 753{,}98\ \Omega$

$X_C = (2\pi \times 1 \times 10^6 \times 150 \times 10^{-12})^{-1} = 1061{,}032\ \Omega$

$Z_1 = 10 + j753{,}98 = 754{,}046 \angle 89{,}24°\ \Omega$ ✓

$Z_2 = 0 - j\ 1061{,}032 = 1061{,}032\ \angle -90°\ \Omega$ ✓

$Z_3 = 1000 + j0 = 1000 \angle 0°\ \Omega$

$$Z_P = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

$$Z_P = \frac{754{,}046\ \angle 89{,}24° \times 1061{,}032\ \angle -90°}{10 - j307{,}052}$$ ✓

$$Z_P = \frac{800066{,}936\ \angle -0{,}76°}{307{,}214\ \angle -88{,}135°}$$ ✓

$Z_P = 2604{,}266\ \angle 88{,}13°\ \Omega$ ✓

$Z_T = Z_P + Z_3 = 84{,}98 + j\ 2602{,}88 + 1000 + j0$ ✓

$Z_T = 1084{,}98 + j\ 2602{,}88$

$Z_T = 2819{,}96\ \angle 67{,}37°\Omega$ ✓

$$I_T = \frac{V_T}{Z_T} = \frac{10\ \angle 0°}{2819{,}96\ \angle 67{,}37°}$$ ✓ $= 3{,}546\ \angle -67{,}37°\ mA$ ✓

(9)

1.2.2 $V_P = I_T Z_P = 3{,}546 \times 10^{-3} \angle -67{,}37 \times 2604{,}266 \angle 88{,}13°$

$V_P = 9.23 \angle 20{,}76° \, V$ ✓

$I_1 = \frac{V_P}{Z_1} = \frac{9.23 \angle 20{,}76°}{754{,}046 \angle 89{,}24°}$ ✓

$I_1 = 12{,}068 \angle -68{,}38° \; mA$ ✓

$I_2 = \frac{V_P}{Z_2} = \frac{9.23 \angle 20{,}76°}{1061{,}032 \; \angle -90°}$ ✓

$I_2 = 8{,}576 \angle 110{,}76° \, mA$ ✓ (5)

[16]

QUESTION 2

2.1
- RC π filter
- LC π filter
- Choke (LC filter)
- RC filter

(Any 2 × 1) (2)

2.2 2.2.1 $V_{r(rms)} = \frac{V_{DC}}{4\sqrt{3} f C R_L}$ ✓

$V_{r(rms)} = \frac{20}{4\sqrt{3} \times 50 \times 0{,}5 \times 1000}$ ✓

$V_{r(rms)} = 115{,}47 \; \mu F$ ✓

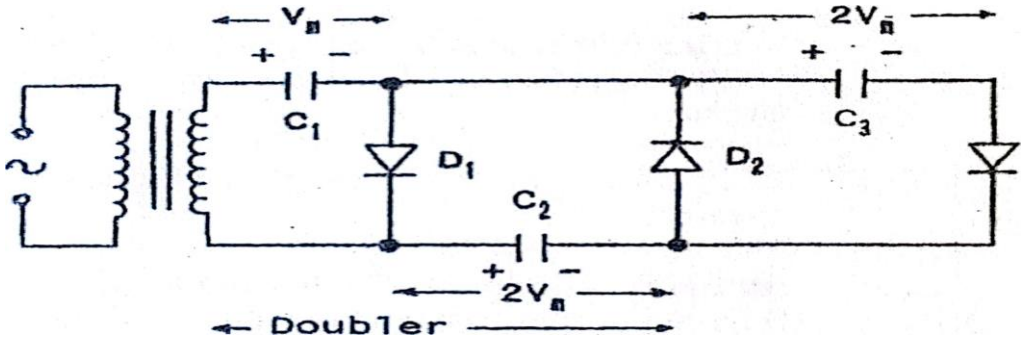
2.2.2 $r = \frac{1}{4\sqrt{3} f C R_L} \times 100$

$r = \frac{1}{4\sqrt{3} \times 50 \times 115{,}47 \times 10^{-6} \times 1000} \times 100$ ✓✓

$r = 2{,}5 \; \%$ ✓

(2 × 3) (6)

2.3



(Half mark for each component including the supply input) (4)

- C_1 charges to Vm through D_1 during positive half cycle
- C_2 charges to 2Vm through D_2 during negative half cycle
- C_3 charges to 2Vm during the positive half cycle as well
- Vm + 2Vm = 3Vm (3)

[15]

QUESTION 3

3.1 3.1.1

$$V_E = I_C R_E$$
$$= 20x10^{-3} \times 150 \checkmark$$
$$= 3\ V \checkmark$$

$$V_E = \frac{1}{10} V_{CC}$$

$$V_{CC} = 3 \times 10 \checkmark$$

$$= 30\ V \checkmark \quad (4)$$

3.1.2

$$R_{B2} = \frac{1}{10} \beta R_E$$

$$= \frac{200 \times 150}{10} \checkmark$$

$$= 3\ k\Omega \checkmark \quad (2)$$

3.1.3

$$R_{B1} = \frac{R_{B2}(V_{CC} - V_B)}{V_B}$$

$$= \frac{3000(30 - 1{,}7)}{1{,}7} \checkmark$$

$$= 49{,}941\ k\Omega \checkmark \quad (2)$$

3.2 3.2.1 $A_i = \left(\frac{h_{fb}}{1+h_{oe}Z_L}\right)\left(\frac{R_E}{R_E+Z_1}\right)\left(\frac{R_C}{R_C+R_L}\right)$

$= \left(\frac{-0,98}{1+(0,8x10^{-6}\times 5194}\right)\left(\frac{1500}{1500+41,267}\right)\left(\frac{6800}{6800+22000}\right)$ ✓

$= (-0,976)(0,973)(0,236)$

$= -0,224$ ✓

$Z_1 = h_{ib} - \frac{h_{fb}h_{rb}Z_L}{1+h_{oe}Z_L}$

$= 40 - \frac{(-0,98)\times 250x10^{-6}\times 5194}{1+(0,8x10^{-6}\times 5194}$ ✓

$= 41,267\ \Omega$ ✓

$Z_L = R_C \quad //R_L$

$= \frac{6800\times 22000}{6800+22000}$

$= 5,194\ k\Omega$ ✓ (5)

3.2.2 $A_V = \frac{-h_{fb}Z_L}{h_{ib}+(h_{ib}h_{ob}-h_{fb}h_{rb})Z_L}$

$= \frac{-(-0,98)\times 5194}{40+(40\times 0,8x10^{-6}-(-0,98)\times 250x10^{-6})\times 5194}$ ✓

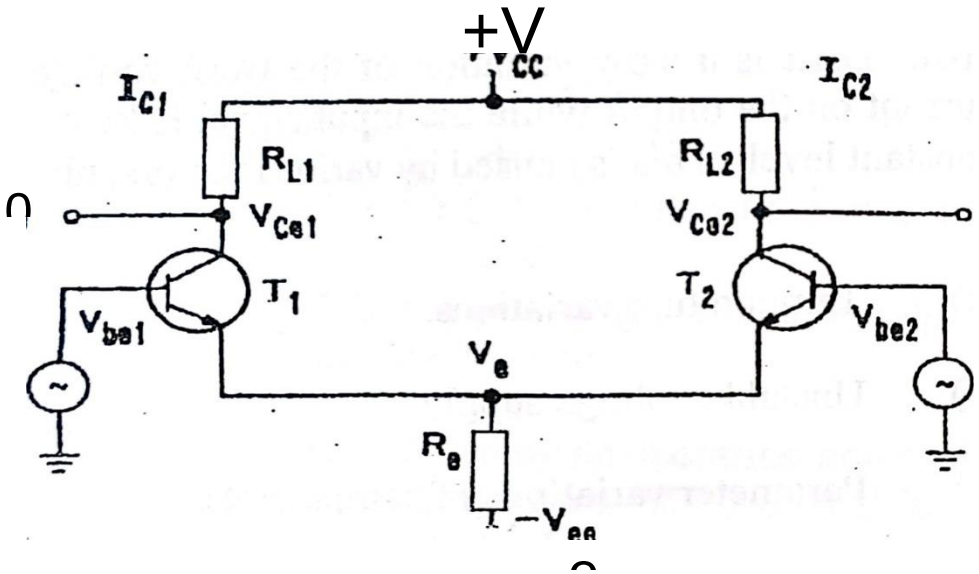
$= 122,83$ ✓ (2)

[15]

QUESTION 4

4.1
- A very high input impedance which causes almost no current load for circuits.
- A very high voltage gain which is suitable to amplify very small signals.
- A very low output impedance which is not affected easily by other signals. (3)

4.2



(1 mark for each transistor, ½ mark each for resistor +Vcc and grounded Vbe) (5)

4.3 4.3.1

$$I_E = \frac{V_E}{R_e} = \frac{16}{2000} = 8\ mA \checkmark$$

$$I_C = \frac{I_E}{2} = \frac{8 \times 10^{-3}}{2} = 4\ mA \checkmark$$

$$R_L = \frac{V_{CC} - V_{CE}}{I_C}$$

$$= \frac{16 - 10}{4 \times 10^{-3}} \checkmark$$

$$= 1{,}5\ k\Omega \checkmark$$ (4)

4.3.2

$$Voltage\ gain = \frac{h_{fe}}{h_{ie}} R_L$$

$$= \frac{50}{800} \times 1500 \checkmark$$

$$= 93{,}75 \checkmark$$ (2)

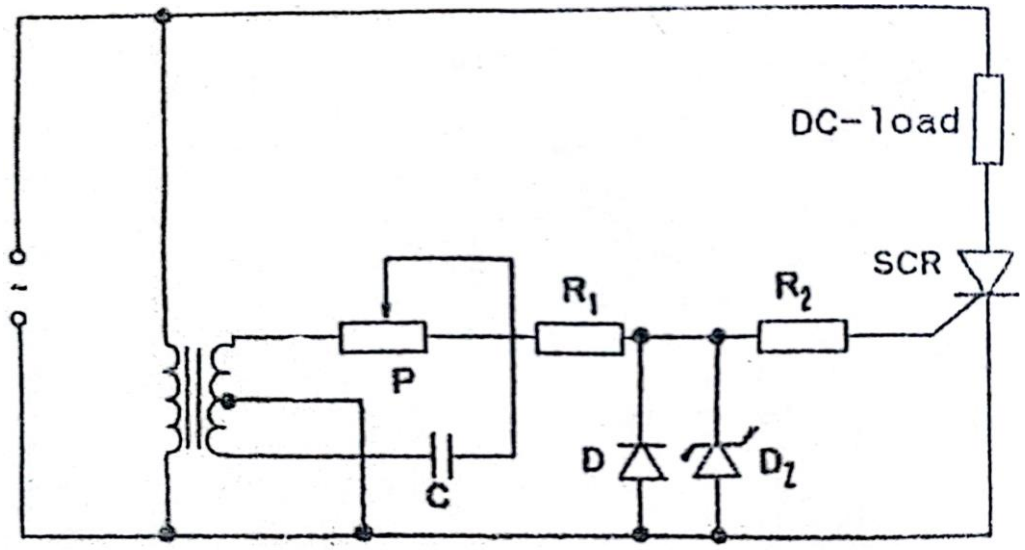
[14]

QUESTION 5

5.1 False
5.2 False
5.3 False
5.4 True

(4 × 1) **[4]**

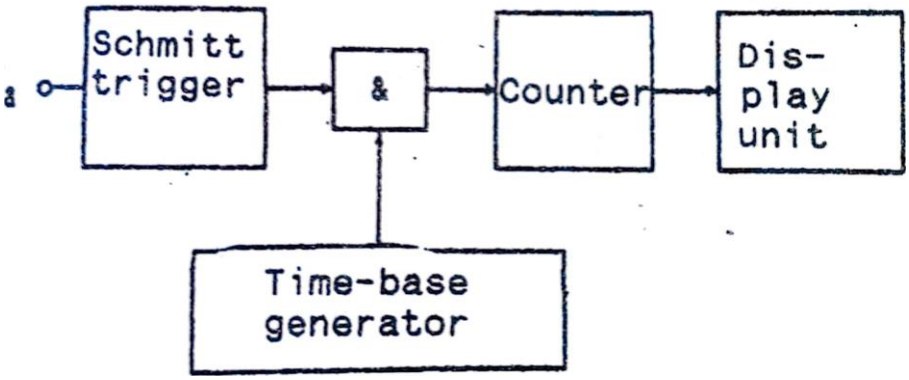
QUESTION 6



(1 mark each for the potentiometer and the centre tap, ½ mark each for all the components including the supply input) **[6]**

QUESTION 7

7.1



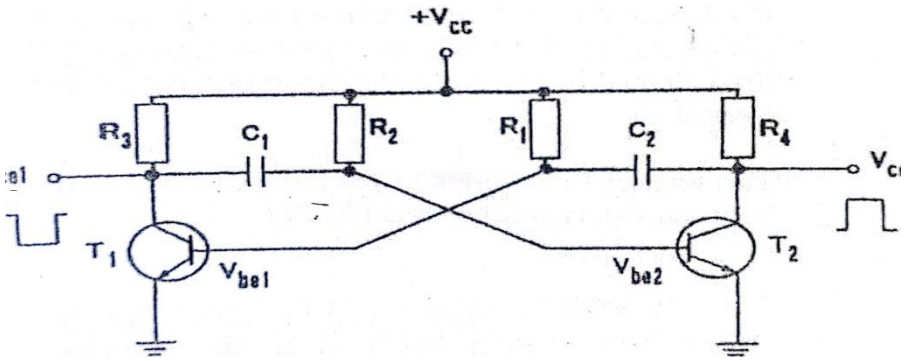
(1 mark for each block) (5)

7.2 To convert analogue input signal to digital signals. (1)
[6]

QUESTION 8

8.1
- Hartely
- Collpits
- RC phase shift
- UJT (4)

8.2



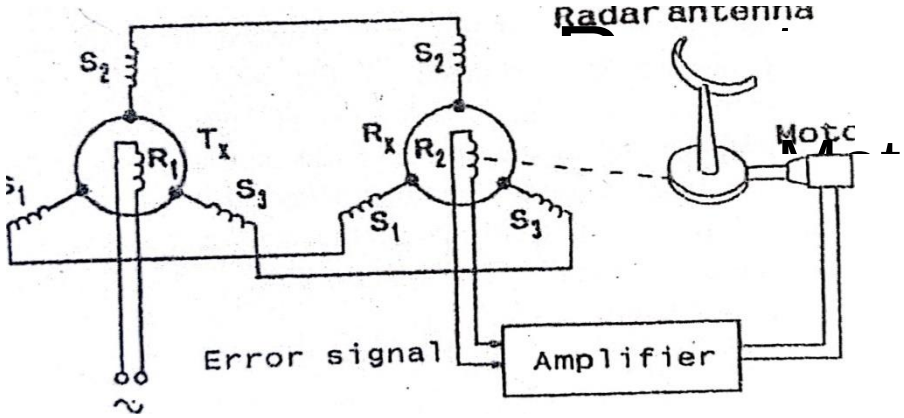
(½ mark for each component) (4)

8.3 Suppose that V_1 is low enough to keep T_1 off.✓ The collector voltage at T_1 is connected to the base of T_2 by means of the voltage divider Rc and R_{b2}.✓ While T_1 is off and its collector voltage high,✓ T_2 will be saturated. The current through T_2 causes voltage drop across Rc and, therefore, determines the emitter potential f T_1.✓ (4)

[12]

QUESTION 9

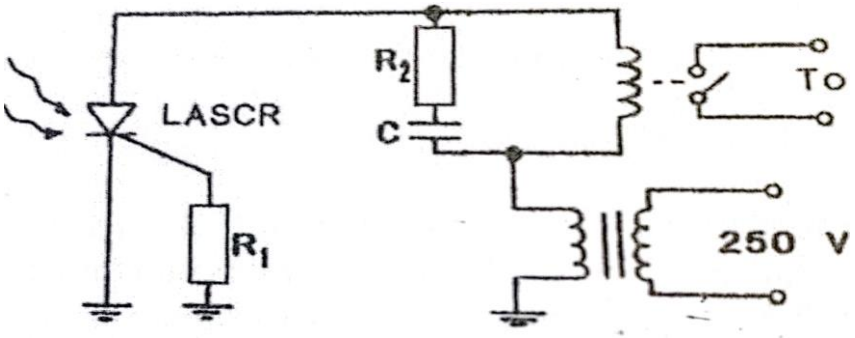
9.1 9.1.1



(1 mark each for transmitter and receiver, one mark each for antenna, motor and amplifier) (5)

9.1.2 The rotor of the receiver will produce an error signal to turn the motor.✓ The antenna also turns the rotor of the receiver at the same angle, θ, so that the error signal becomes zero.✓ (2)

9.2 9.2.1



(½ mark for each component) (3)

9.2.2
- AND function
- OR function (2)

[12]

TOTAL: 100

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